# Lab 02: Model Fitting & SfM

Computer Vision, ETH Zurich, HS2026

## 1 Structure from Motion

The reconstruction used the ten supplied images and feature matches. Images 0003 and 0004 were used for initialization, with image 0003 defining the world frame. The input images, keypoints and matches were checked using the provided visualization functions.

### 1.1 Essential matrix and initial pose

Pixel coordinates were converted to homogeneous rays using the inverse calibration matrix. For the constraint in the assignment,

$$x_1^{\mathrm{T}} E\, x_2 = 0,$$

each DLT row contains the entries of $x_1 x_2^{\mathrm{T}}$ in row-major order. The last right singular vector gives the initial estimate. Its singular values were replaced by (1, 1, 0) to enforce the essential-matrix constraints.

The convention matters here: the theory exercise writes $x_2^{\mathrm{T}} E\, x_1 = 0$ [3]. The coding task therefore uses the transpose of that matrix. The supplied decomposition maps camera 2 to camera 1, whereas the stored poses map world coordinates to camera coordinates. With camera 1 fixed to $[I \mid 0]$, the second pose is

$$R_2 = R_{21}^{\mathrm{T}}, \quad t_2 = -R_{21}^{\mathrm{T}} t_{21}.$$

All four decompositions were tested by triangulation. Their positive-depth counts were [5765, 0, 0, 0], so the first candidate was selected in this run. All 5765 initial matches produced points in front of both cameras.

### 1.2 Triangulation and map extension

After linear triangulation, non-finite points and points with non-positive depth in either camera were removed. The same masks were applied to the keypoint indices to preserve the 2D-3D correspondences.

For each new image, the 2D correspondences were normalized before applying the provided absolute-pose DLT. New points were triangulated against every registered image. A shared feature reused its pending point index across pairs; this avoided creating several points for the same track. Before adding a batch to the reconstruction, the local indices were offset by the previous number of points.

Seven numerical tests passed, including known-pose recovery, depth filtering and correspondence indexing. The completed parts of the framework were retained. No bundle adjustment was performed.

## 1.3 Reconstruction results

All ten cameras were registered. The final map contains 13,588 points and 61,848 stored image observations. Each point is observed in at least two images.

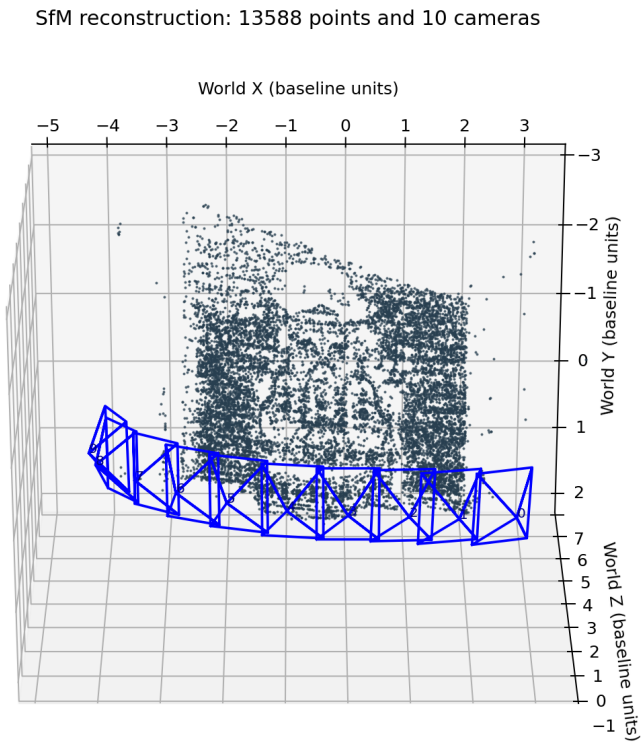



**Figure 1.** Reconstructed scene and camera poses. Blue frusta indicate the cameras; labels 0-9 correspond to the input image numbers. The initial baseline is one unit. The axis limits follow the supplied percentile-based display settings; points outside the displayed range are retained in the reconstruction.

**Table 1.** Reprojection error over all stored observations.

| Median (px) | RMSE (px) | 95th percentile (px) |
|---|---|---|
| 3.754 | 18.879 | 37.274 |

The fountain and wall are visible in the point cloud, and the camera sequence follows the viewpoints in the input images. All stored observations have positive depth. However, the large difference between the median error and RMSE shows that some observations are reconstructed much less accurately.

Image 0009 has a median reprojection error of 54.302 px, compared with 0.437 px for image 0003. The increasing error in later views is consistent with error accumulation in the sequential DLT pipeline, although this run does not isolate its cause. Positive depth alone is insufficient to reject inaccurate points. Joint refinement of the cameras and points would be a natural next step.

## 2 Model fitting

The supplied generator produced 500 observations of $y = x + 10$, including 50 injected outliers. Both random seeds were set to 0. Least squares was solved using the design matrix $[x, 1]$. RANSAC ran for 300 iterations with five observations per sample and an inlier threshold of 1. The distance was

$$d_i = |k\,x_i - y_i + b| \,/\, (\,k^2 + 1\,)^{1/2}.$$

A candidate replaced the current model only when it had more inliers. The final model was not refitted to the full inlier set.

**Table 2.** Ground-truth and estimated line parameters.

| Method | k | b |
|---|---|---|
| Ground truth | 1.000000 | 10.000000 |
| Least squares | 0.615966 | 8.961727 |
| RANSAC | 0.964389 | 9.982921 |

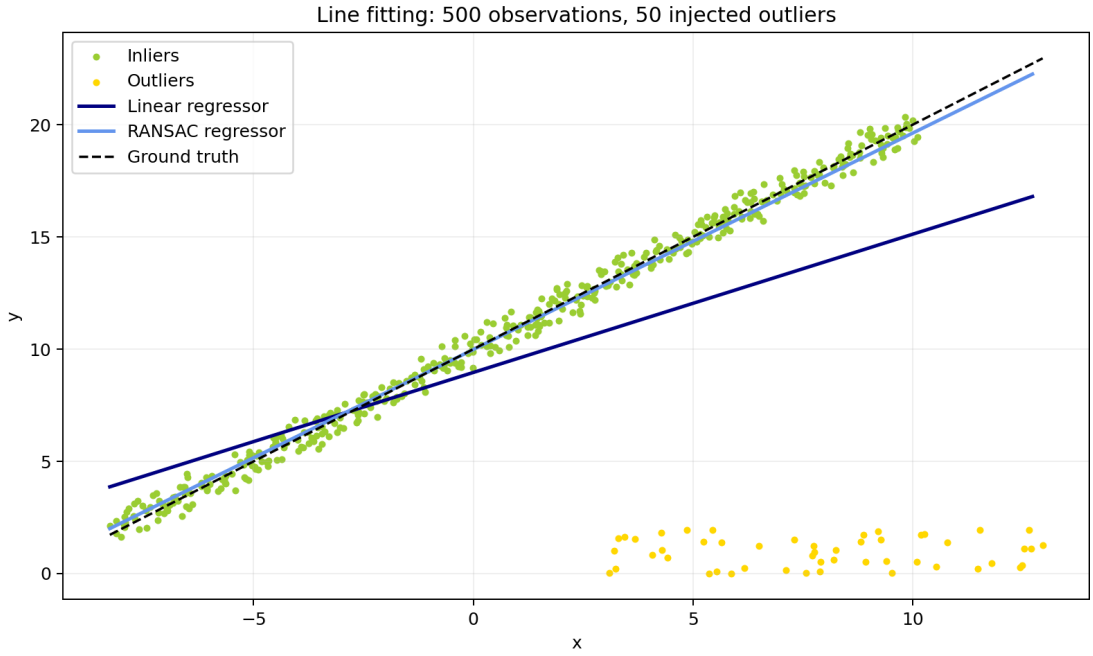



**Figure 2.** Least-squares and RANSAC fits. Green points are RANSAC inliers; yellow points are rejected observations. The dashed line shows the ground truth.

Least squares is pulled towards the outliers at high $x$ and low $y$, reducing the slope to 0.616. RANSAC rejected all 50 injected outliers and retained the other 450 observations. Its estimate is closer to the ground truth, but the slope still differs by about 0.036. This is plausible for a model fitted to only five noisy observations without a final consensus-set refit.

### References


[1] Lab 02 Assignment - Model Fitting & SfM, HS2026, pp. 1-3.

[2] Lab 2 coding slides, HS2026, slides 7-13.

[3] Exam Prep - Model Fitting & SfM, questions and solutions, HS2026.